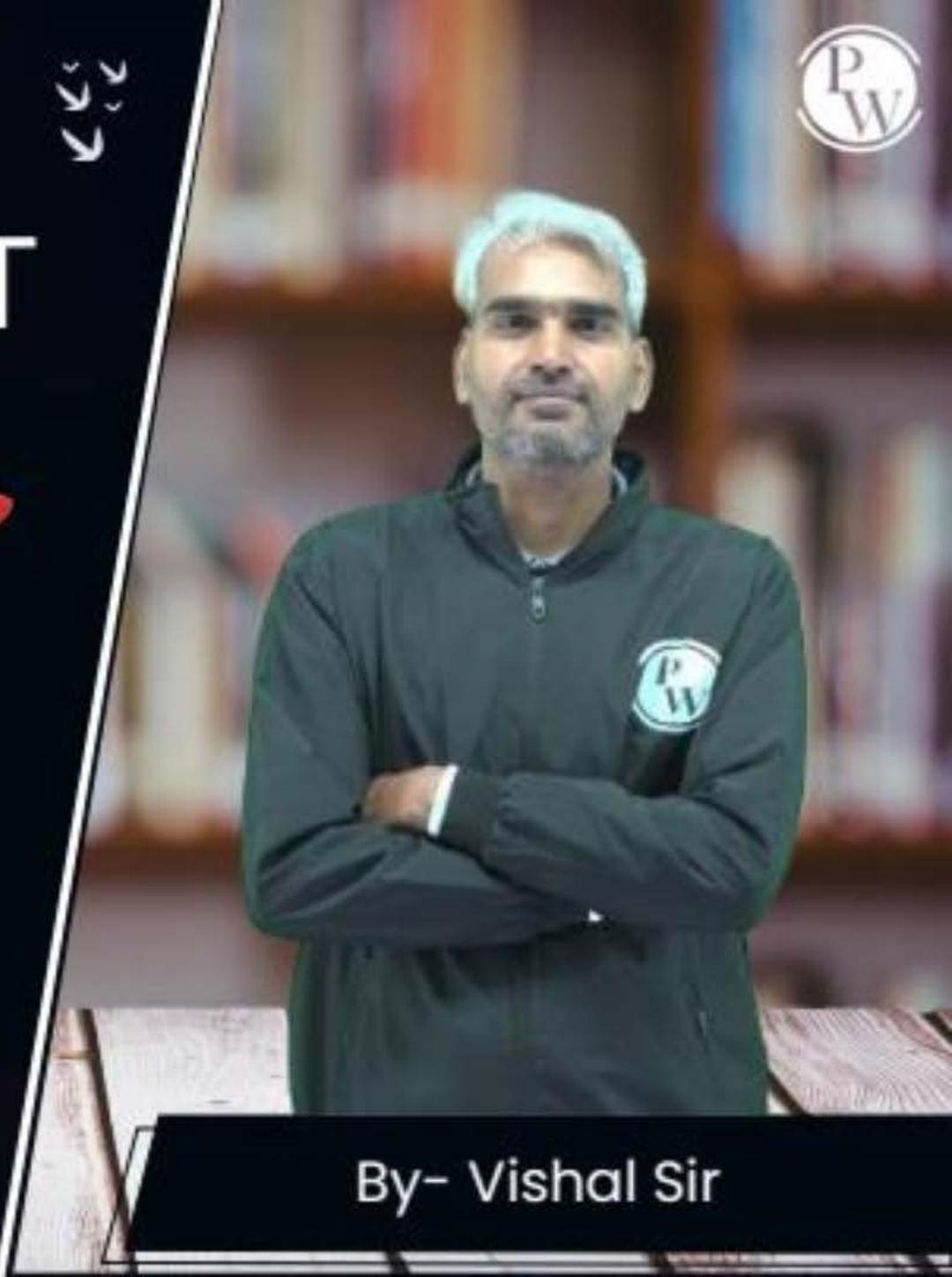
Computer Science & IT



Database Management System

Transaction & concurrency control

Lecture No. 09



By- Vishal Sir

Recap of Previous Lecture



- Topic: Practice questions on conflict serializability
- Topic: View serializability conditions

Slide

Topics to be Covered



Topic	Practice questions on View serializable schedule

Slide



Topic : View serializable schedule

- If given schedule is view equivalent to at least one of the serial schedule , then it is called a view serializable schedule.

Converse of the statement is also true



Topic : View equivalent condition

- Consider two schedules S1 and S2
- Schedules S1 and S2 are called view equivalent if the following three conditions hold true for them –
  - Condition-01 : For each data item X, if transaction Ti reads X from the initial database in schedule S1, then in schedule S2 also, Ti must perform the [redacted] read of X from the initial database.

Thumb Rule

- "Initial readers must be same for all the data items".



Topic : View equivalent condition

- Condition-02 :

  If transaction Ti reads the value of data item X updated by the transaction Tj in schedule S1, then in schedule S2 also, transaction Ti must read the value of data item X updated by the transaction Tj.

  Thumb Rule

  - "Write-read sequence must be same.".



Topic : View equivalent condition

- Condition-03 :

For each data item X, if X is finally updated by transaction Ti in schedule S1, then in schedule S2 also, X must be finally updated by transaction Ti.

Thumb Rule

- "Final writers must be same for all the data items".

- For two schedules to be view equivalent all three conditions must be satisfied

* Check whether the following schedule is view serializable schedule or not? If view serializable schedule, then identify all view equivalent serial schedules.

S_1

T_1	T_2	T_3
$W_1(A)$		
	$R_2(B)$	
	$R_2(A)$	
	$W_2(B)$	
$W_1(B)$		
		$W_3(B)$

* Check whether the following schedule is view serializable schedule or not? If view serializable schedule, then identify all view equivalent serial schedules.

S1

T_1	T_2	T_3
$W_1(A)$		
	$R_2(B)$	
	$R_2(A)$	
	$W_2(B)$	
$W_1(B)$		
		$W_3(B)$

✱1✱ Identify the constraints w.r.t Read from initial db.

Data item	Transactions that reads the dataitem from initial database	Other transactions that update the dataitem	Constraints
A	—	T_1	No Constraint
B	T_2	$\{T_1, T_3\}$	$T_2 \rightarrow \{T_1, T_3\} \equiv T_2 \rightarrow T_1$ & $T_2 \rightarrow T_3$

* Check whether the following schedule is view serializable schedule or not? If view serializable schedule, then identify all view equivalent serial schedules.

S1

T_1	T_2	T_3
$W_1(A)$		
	$R_2(B)$	
	$R_2(A)$	
	$W_2(B)$	
$W_1(B)$		
		$W_3(B)$

2 Identify the Constraints w.r.t. WR sequence:-

Let T_i writes dataitem X in the schedule & T_j reads the value written by T_i and T_k is some other writer of dataitem X.

then in equivalent serial schedule T_i must execute before T_j i.e.; $T_i \longrightarrow T_j$

and no writer of X must be allowed to execute in between T_i & T_j → i.e.: $T_k = \phi$ between T_i & T_j

* Check whether the following schedule is View Serializable Schedule or not? If View Serializable Schedule, then identify all View equivalent Serial schedules.

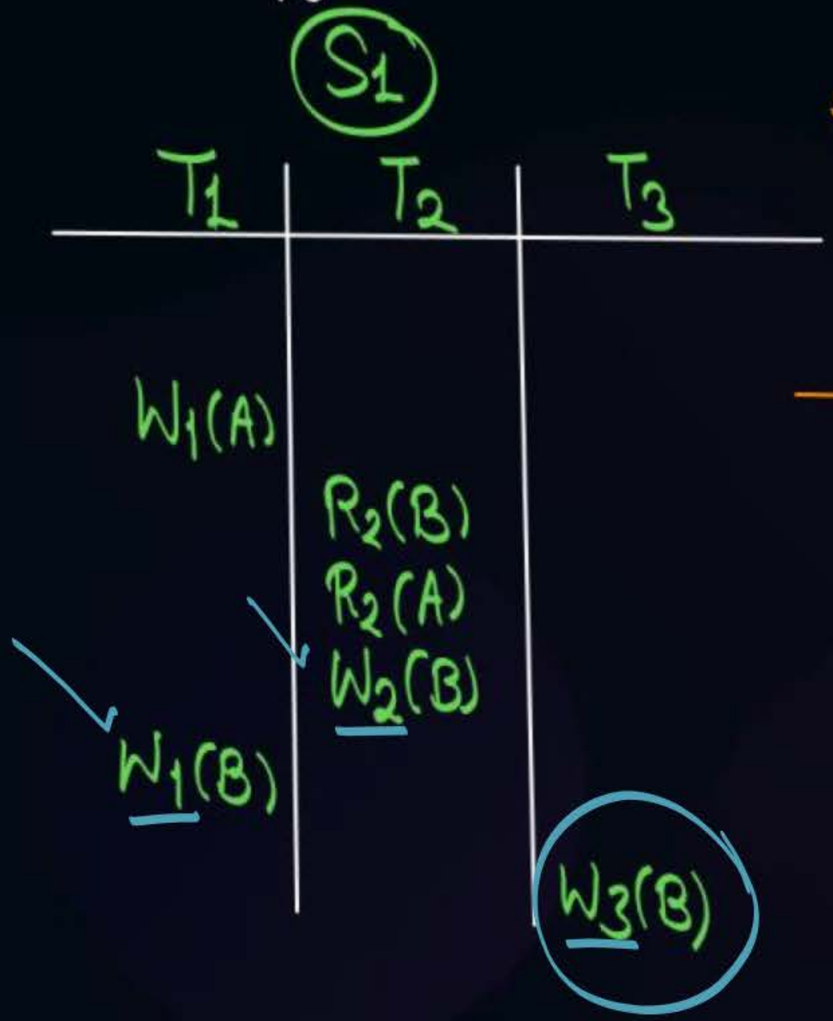
(S1)

T_1	T_2	T_3
$W_1(A)$		
	$R_2(B)$	
	$R_2(A)$	
	$W_2(B)$	
$W_1(B)$		
		$W_3(B)$

(2) Identify Constraints w.r.t Write-Read Sequence :-

Dataitem	Identification of Constraints
A:	T_1: writes A, and T_2 read the value written by T_1 No other write of A $\therefore T_k$ is already $\emptyset$ $\therefore$ Constraint is $T_1 \rightarrow T_2$
B	No write-Read seq. w.r.t. dataitem B $\therefore$ No Constraint

* Check whether the following schedule is view serializable schedule or not? If view serializable schedule, then identify all view equivalent serial schedules.



3 Identify the Constraints w.r.t. final write

Data item	Transaction with final update	other writers	Constraints
A	T_1	—	No Constraints
B	T_3	$\{T_1, T_2\}$	$\{T_1, T_2\} \rightarrow T_3 \equiv T_1 \rightarrow T_3$, $T_2 \rightarrow T_3$

Note:- Once we have identified the constraints w.r.t. all three conditions, We will Construct the dependency graph w.r.t. overall Constraints.

- If dependency graph is acyclic then given schedule is a "view serializable schedule" and hence a serializable schedule
- If dependency graph is cyclic then given schedule is not a "view serializable schedule" and hence not a serializable schedule

- A schedule is serializable schedule
  if and only if schedule is view serializable schedule.

* Check whether the following schedule is View Serializable Schedule or not? If View Serializable Schedule, then identify all view equivalent Serial schedules.

S_1

T_1	T_2	T_3
$W_1(A)$		
	$R_2(B)$	
	$R_2(A)$	
	$W_2(B)$	
$W_1(B)$		
		$W_3(B)$

Overall constraints

$\{ T_2 \rightarrow T_1,\ T_2 \rightarrow T_3 \}$

$\{ T_1 \rightarrow T_2 \}$

$\{ T_1 \rightarrow T_3,\ T_2 \rightarrow T_3 \}$

$\Rightarrow$ $T_2 \rightarrow T_1$, $T_2 \rightarrow T_3$, $T_1 \rightarrow T_2$, $T_1 \rightarrow T_3$

∴ Not V.S.S. & hence Not Serializable

In the set of overall constraints if we find two constraints s.t. $T_i \rightarrow T_j$ & $T_j \rightarrow T_i$, then dependency graph is guaranteed to be Cyclic. Hence Schedule is not a View Serializable Schedule

* Check whether the following schedule is View Serializable Schedule or not? If View Serializable Schedule, then identify all view equivalent Serial schedules.

S_1

T_1	T_2	T_3
$W_1(A)$		
	$R_2(B)$	
	$R_2(A)$	
	$W_2(B)$	
$W_1(B)$		
		$W_3(B)$

Overall constraints

$$\left\{\begin{array}{l} T_2 \rightarrow T_1 \\ T_2 \rightarrow T_3 \end{array}\right.$$

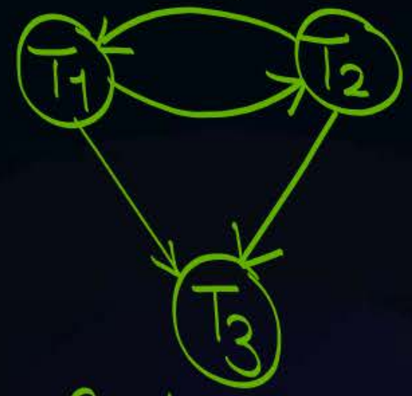
$$\left\{ T_1 \rightarrow T_2 \right.$$

$$\left\{\begin{array}{l} T_1 \rightarrow T_3 \\ T_2 \rightarrow T_3 \end{array}\right.$$

$\Rightarrow$

$T_2 \rightarrow T_1$
$T_2 \rightarrow T_3$
$T_1 \rightarrow T_2$
$T_1 \rightarrow T_3$

Dependency graph



Cyclic
∴ Not a V.S.S.

Hence, Not a Serializable Schedule

Q: * Check whether the following schedule is view serializable schedule or not? If view serializable schedule, then identify all view equivalent serial schedules.

S_2

T_1	T_2	T_3
		$R_3(B)$
$W_1(A)$		
	$R_2(B)$	
	$R_2(A)$	
	$W_2(B)$	
$W_1(B)$		
		$W_3(B)$

Q: * Check whether the following schedule is view serializable schedule or not? If view serializable schedule, then identify all view equivalent serial schedules.

S_2

T_1	T_2	T_3
		$R_3(B)$
$W_1(A)$		
	$R_2(B)$	
	$R_2(A)$	
	$W_2(B)$	
$W_1(B)$		
		$W_3(B)$

① Identify the Constraints w.r.t. Read from initial database

Data item	Transaction Reading from initial database	other Writers	Constraints
A	—	T_1	No Constraint
B	T_2, T_3	T_1, T_3	$T_2 \to \{T_1, T_3\} \equiv T_2 \to T_1$, $T_2 \to T_3$
		T_1, T_2	$T_3 \to \{T_1, T_2\} \equiv T_3 \to T_1$, $T_3 \to T_2$

Cyclic

Q11: * Check whether the following schedule is view serializable schedule or not? If view serializable schedule, then identify all view equivalent serial schedules.

S_2

T_1	T_2	T_3
		$R_3(B)$
$W_1(A)$		
	$R_2(B)$	
	$R_2(A)$	
	$W_2(B)$	
$W_1(B)$		
		$W_3(B)$

* Constraints w.r.t. Write-Read sequence
  &
  Constraints w.r.t. Final update
  are exactly same as previous question

Q:→ Check whether the following schedule is a V.S.S. or not? If V.S.S. then identify all serial schedules that are view equivalent to given schedule

S_3

T_1	T_2
$W_1(A)$	
	$R_2(A)$
	$R_2(B)$
$W_1(B)$	

① Constraints w.r.t. Read from initial database

Dataitem → A : No Constraint

Dataitem → B : $T_2 \rightarrow T_1$

② Constraints w.r.t. Write-Read sequence

Dataitem → A : $T_1 \rightarrow T_2$

Dataitem → B : No WR Seq. wrt B

③ Constraints w.r.t. final update

Dataitem → A : No Constraint

Dataitem → B : No Constraint

$T_1 \rightarrow T_2$ & $T_2 \rightarrow T_1$

∴ Cyclic dependency graph

Hence Not a V.S.S.

∴ Not a Serializable schedule

Q:- Check whether the following schedule is a V.S.S. or not? If V.S.S. then identify all serial schedules that are view equivalent to given schedule

S4

T_1	T_2	T_3
	$R_2(B)$	
	$R_2(A)$	
$W_1(A)$		
		$W_1(A)$
	$W_1(A)$	

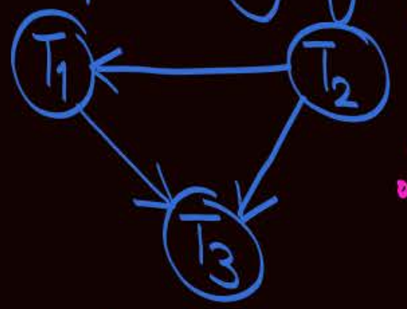
① Constraints w.r.t. Read from initial database
A: $T_2 \rightarrow \{T_1, T_3\} \equiv T_2 \rightarrow T_1$, $T_2 \rightarrow T_3$
B: —

② WR seq.
A: $T_2 \rightarrow T_1$
B: —

③ Final write
A: $\{T_1, T_2\} \rightarrow T_3 \equiv T_1 \rightarrow T_3$, $T_2 \rightarrow T_3$
B: —

Overall Constraints = $T_2 \rightarrow T_1$, $T_2 \rightarrow T_3$, $T_1 \rightarrow T_3$

Dependency graph



Acyclic ∴ V.S.S.

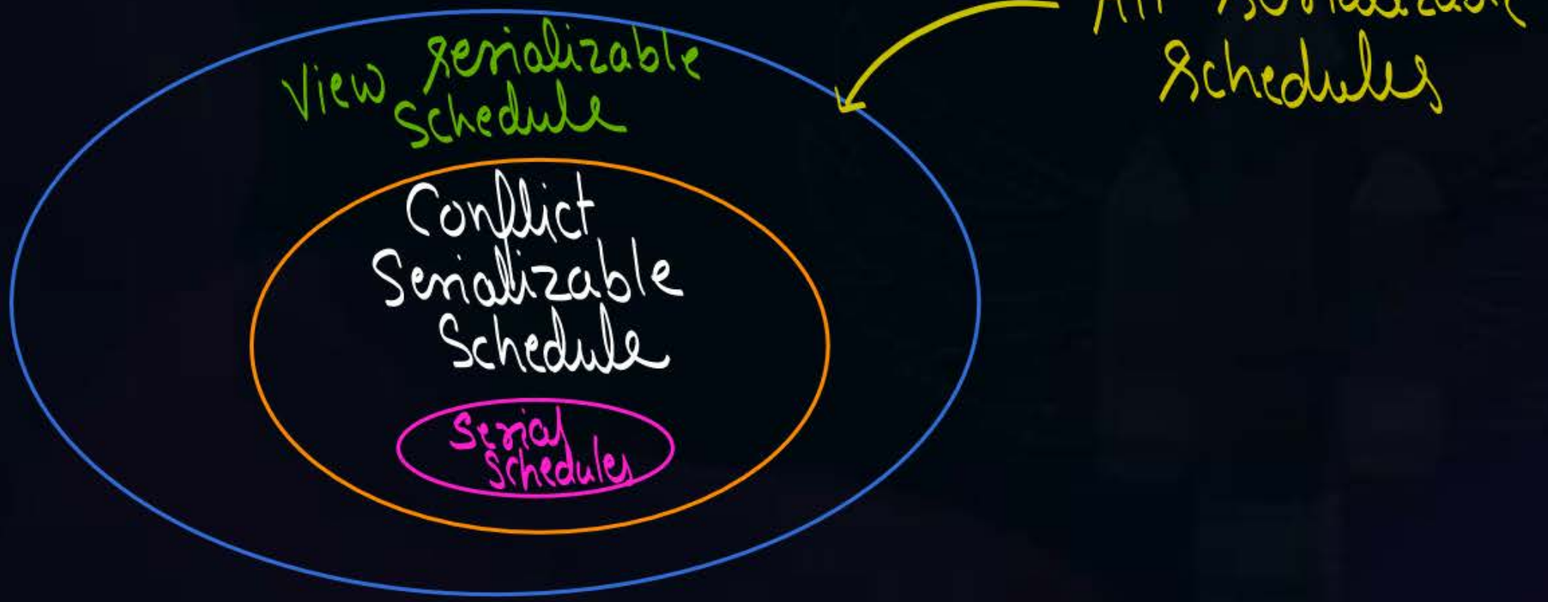
and view equivalent serial schedule is

$T_2 \rightarrow T_1 \rightarrow T_3$



Topic : Conflict serializable & View serializable schedule

* Every Conflict serializable schedule is a View serializable schedule, but Every View serializable schedule need not be a Conflict serializable schedule



#Q. Check whether the schedule is view serializable schedule or not?

If view serializable schedule then identify all view equivalent serial schedules.

T1	T2	T3	T4
R(A)			
	R(A)		
		R(A)	
			R(A)
W(B)			
	W(B)		
		W(B)	
			W(B)

① Read from initial database

A: —

B: —

② Write Read Seq

A —

B —

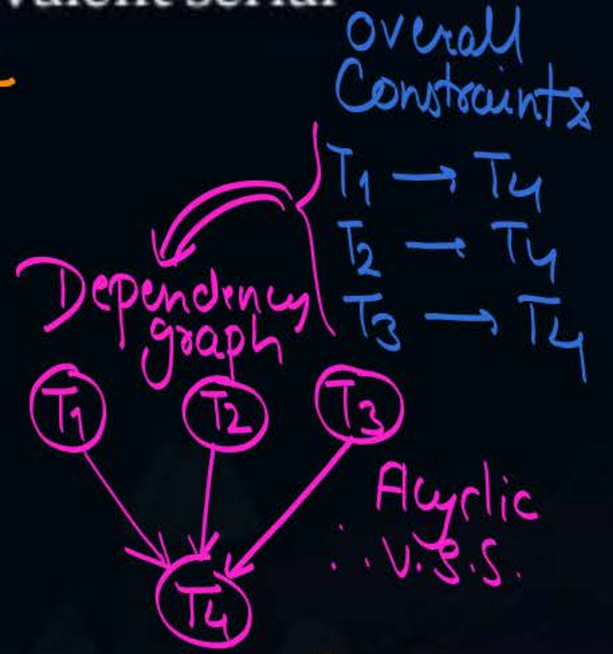
③ Final Update

A —

B $(T_1, T_2, T_3) \rightarrow T_4$

T_1, T_2 & T_3 can appear before T_4 in $3! = 6$ ways

Hence '6' View equivalent serial schedules



$T_1 \rightarrow T_2 \rightarrow T_3 \rightarrow T_4$

$T_1 \rightarrow T_3 \rightarrow T_2 \rightarrow T_4$

$T_2 \rightarrow T_1 \rightarrow T_3 \rightarrow T_4$

$T_2 \rightarrow T_3 \rightarrow T_1 \rightarrow T_4$

$T_3 \rightarrow T_1 \rightarrow T_2 \rightarrow T_4$

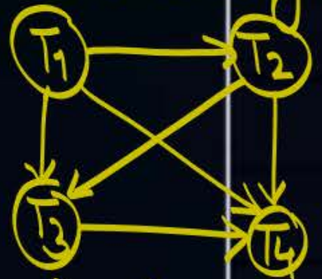
$T_3 \rightarrow T_2 \rightarrow T_1 \rightarrow T_4$

#Q. How many serial schedules are there over the transactions of schedule S that are view equivalent to schedule S but not Conflict equivalent to schedule S.

"S"

T1	T2	T3	T4
R(A)			
	R(A)		
		R(A)	
			R(A)
W(B)			
	W(B)		
		W(B)	
			W(B)

Precedence graph



Acyclic
∴ C.S.S.

Topological ordering

$T_1 \to T_2 \to T_3 \to T_4$

⇒ Schedule S is view equivalent to '6' serial schedules

& Conflict equivalent to only '1' serial schedule

∴ # Serial schedules that are View equivalent to S but not Conflict equivalent to S = 6 − 1 = 5

and they are, $T_1 \to T_3 \to T_2 \to T_4$, $T_2 \to T_1 \to T_3 \to T_4$

$T_2 \to T_3 \to T_1 \to T_4$, $T_3 \to T_1 \to T_2 \to T_4$

& $T_3 \to T_2 \to T_1 \to T_4$

#Q. Check whether the schedule is view serializable schedule or not?

If view serializable schedule then identify all view equivalent serial schedules.

T1	T2	T3
R(A)		
	R(B)	
	W(B)	
		R(A)
		W(A)
R(B)		
		R(B)
	R(A)	
	W(A)	

① Read from initial db

A: $T_1 \rightarrow \{T_2, T_3\} \equiv T_1 \rightarrow T_2$, $T_1 \rightarrow T_3$, $T_3 \rightarrow T_2$

B: ——

② WR Seq

A: $T_3 \rightarrow T_2$

B: $T_2 \rightarrow T_1$, $T_2 \rightarrow T_3$

③ final update

A: $T_3 \rightarrow T_2$

B: ——

Cycle ∴ Not V.S.S.

Hence, not a Serializable Schedule



2 mins Summary

Topic Practice questions on View serializable schedule

Slide

Next Class :.

Today at 1:30 PM

THANK - YOU

Slide